



SECTION A

General Unit Information:

Unit Name	Software For Enterprise
Unit Code	CIS046-3
Level	6
Credit Value	30
Location(s) of Delivery	On Campus
Period(s) of Delivery	15 weeks
Pre-requisites or Restrictions	N/A
Aims and Relevance	Modern Software lives in the context of other software. Software Engineers massively interact with each other, be it in local teams, via Question and Answer sites (such as “Stack Overflow”) or in professional version control hosting services (such as “GitHub”). Interoperability, traceability and the professional definition of application interfaces to enable others to use or contribute to the software become more and more relevant. This unit aims to prepare you for this collective development of software.
Syllabus Content	• Professional organizations, such as the Object Management Group (OMG), the World Wide Web Consortium (W3C) and the Internet Society (ISOC) • Version control and collaborative platforms • Application interfaces, Interoperability and the role of JSON and XML • Service Oriented Architectures • Event driven programming and architectures, events and user interfaces, events and embedded systems • Programming languages for the web: the role of JavaScript, PHP and others • Web Services and related concepts • Commercial aspects and economy of service based architectures • Security and identity in distributed software systems • Origin, purpose and maintenance of protocols and Internet standards • Synchronous and asynchronous communication

Learning Outcomes

On completion of this unit you should be able to

1. Demonstrate the following knowledge and understanding

Differentiate and critically question the context and underpinning principles of collectively developing software to industry standard

2. Demonstrate the following skills and abilities

Develop, implement and critically appraise software solutions that are tailored against specific professional requirements

PSRB Outcomes and/or Apprenticeship Standard

N/A

Summary Learning Hours

	Scheduled	Guided	Independent	Autonomous	Placement	Total
Hours	48	60	84	108	0	300
Percentage	16%	20%	28%	36%	0%	100%

Approach to Learning

All five dimensions to realistic learning (meaningful, active, reflective, challenging and collaborative) are addressed, in particular the following elements are included:

- Guided and unsupervised research using electronic and paper based resources.
- Students are exposed to a variety of software architectures in the context of a modern high-profile operating system.
- Lectures that introduce the fundamental principles and prepare the theoretical foundation for the practical work.

The practical classes will let the students consolidate what they have learnt from the lectures and obtain some useful programming skills in the various architectures. The practical sessions are comprised of three elements: demonstrations by the tutor; attempts on own implementation by the student; discussion between student and tutor about the software they produced.

Assessment Summary

No	Assessment Method	Learning Outcome(s)	Weight	Submission Week	Length of Assessment
1	Coursework - Artefact	1 and 2	30	10	1500 words
2	Ex	1 and 2	70	13	3 hours

Learner Development

This Unit particularly focuses on the development of your abilities in the following areas:

Enquiry	The practical classes will let the students consolidate what they have learnt from the lectures and obtain some useful programming skills in the various architectures.
Contextual Understanding, including where employer based	Lectures that introduce the fundamental principles and prepare the theoretical foundation for the practical work.
Collaboration	You will work with your peers on some of the assignment tasks.

Enterprise	To help with the development of this you will develop software based on your own ideas and self-assess the results in view of your own experience of peers using your product.
Assessment 1 Details	You will submit an artefact as specified in the format outlined in the detailed assessment brief for the unit. This will involve a solution that utilises enterprise standard software development skills.
Assessment 2 Details	You will undertake a closed-book examination of 3 hours duration relating to core technologies and techniques for developing enterprise level software systems.
Assessment 3 Details	
Assessment 4 Details	

Threshold Expectations

In order to pass Assessment 1 you will need to :

- Comment your software so that the logical working structure is clear to a software developer
- Define application interfaces for your software
- Contextualize your software within a version control system

In order to pass Assessment 2 you will need to:

- Select and/or specify software solutions to a given case study
- Identify application areas where a given software standard or architecture is relevant
- Question and evaluate the origin, purpose and context of software standards
- Devise strategies to enable interoperability
- Discriminate between different types of authentication in distributed software

In order to pass Assessment 3 you will need to:

In order to pass Assessment 4 you will need to:

Section B

Recommended Reading

Core Text - this unit is supported by the following core text:

There is no Core Text. Relevant links are provided on BREO

Guided Reading

The following is expected reading for this unit. Details of what to read and when will be provide in the BREO site.

See <https://rl.talis.com/3/bedfordshire/lists/F0FE7114-BBD7-42A6-1E45-AD2AC34EA050.html>

Independent Study

To receive high grades you will need to demonstrate your wider reading. The following resources provide useful background reading for the material in this unit. This is not an exhaustive list and students should read widely from the variety of journals available in the Learning Resources Centre.

See <https://rl.talis.com/3/bedfordshire/lists/F0FE7114-BBD7-42A6-1E45-AD2AC34EA050.html>

Equality Impact Assessment

Question	Y/N/NA	Additional anticipatory adjustments/actions if necessary
Learning materials will be made available in advance of sessions for students to adapt as appropriate?	Y	
The approach to teaching and learning is sufficiently flexible to enable all students to succeed?	Y	
The approach to group work takes account of the needs of students with disabilities and from diverse backgrounds?	Y	
The approach to practical work takes account of the needs of students with disabilities?	Y	The practical work in this unit relies on intensive use of computing and engineering equipment and so if you have difficulty accessing these you should discuss this with the Disability Advice Team in conjunction with the course team at the outset to ensure that appropriate support is in place.
Students with a protected characteristic* have an equal opportunity to achieve the learning outcomes?	Y	
The assessment tasks provide all students with an equal opportunity to succeed?	Y	

Any other aspects of the unit that might pose potential challenges from an equality or diversity perspective have been considered?	Y	
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* Age, Gender reassignment, Marriage and civil partnership, Pregnancy and maternity, Race, Religion and belief, Sex, Sexual orientation

Section C

Administrative Information – Faculty completion	
Faculty	Creative Arts Technologies and Science
Portfolio	CU-COM
School/Department	School of Computer Science and Technology
HECOS Code(s)	100374
Unit Co-ordinator	Marc Conrad
Apprenticeship ?	No

Courses that this unit is delivered on
BSc (Hons) Software Engineering